

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

*Authors: Jacob Devlin, Ming-Wei Chang, Kenton Lee, Kristina Toutanova
(Google AI Language, 2019)*

Student Name: Feliciann Elliot **USI:** 1022055

Motivation and Research Problem

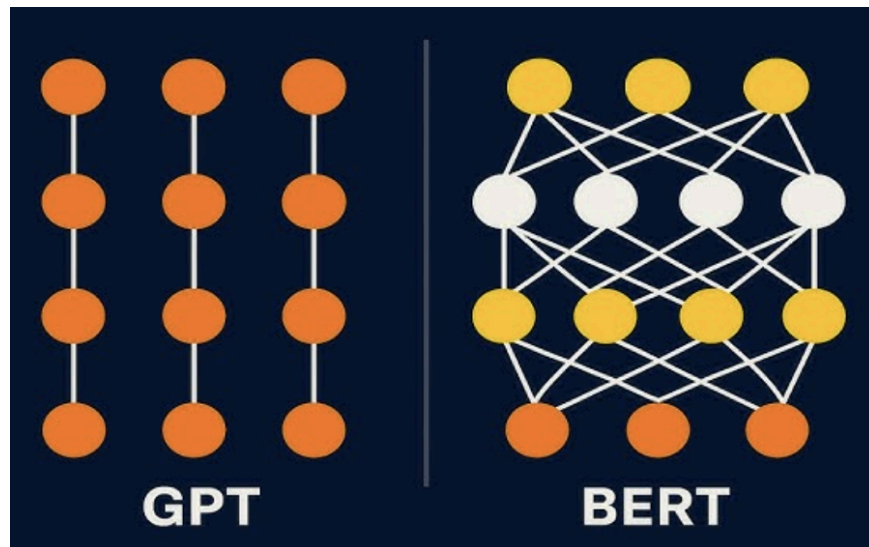
Why BERT Was Needed

- Previous models (GPT, ELMo) read text only one way — left → right or right → left.
- Couldn't understand both sides of context.

Example:

"The bank near the river" vs "The bank approved the loan"

➡ BERT reads both directions simultaneously (bidirectional), solving this limitation.



Historical Context & Significance

From Word Embeddings to Deep Context

- **word2vec, GloVe** → one meaning per word (no context)
- **ELMo** → contextual embeddings
- **GPT** → fine-tuning, but unidirectional
- **BERT** → contextual + bidirectional

Significance:

State-of-the-art on **11 benchmarks** (GLUE 80.5, MNLI 86.7, SQuAD 93.2 F1)

Methodology Overview

Two-Phase Approach: Pre-training + Fine-tuning

Pre-training:

- Data: BooksCorpus (800M words) + Wikipedia (2.5B words)
- Tasks: Masked Language Modeling (MLM) & Next Sentence Prediction (NSP)

Fine-tuning:

- Adapt model to downstream tasks (e.g., QA, sentiment)
- 💡 Analogy: Teach general language → then test on specific subjects.

Architecture and Algorithm

Transformer Encoder Structure

- Based on *Vaswani et al., 2017*
- Uses **self-attention** (every word attends to every other)
- **BERT_BASE**: 12 layers, 768 hidden units, 12 heads (110M params)
- **BERT_LARGE**: 24 layers, 1024 hidden units, 16 heads (340M params)

 Bidirectionality = Training with masked words, not fixed order.

Input Representation

How Sentences Enter the Model

- Supports single or paired sentences
- Token embedding = Token + Segment + Position

Special Tokens:

- [CLS] → classification token
- [SEP] → separates sentences

Example:

[CLS] I love AI [SEP] It's revolutionary [SEP]

- ✓ Unified input makes cross-task fine-tuning easy.

Pre-training Task 1 – Masked Language Modeling (MLM)

- Randomly mask 15% of words
 - 80% → [MASK]
 - 10% → random word
 - 10% → unchanged

Task: Predict the masked words

Example: “The cat sat on the [MASK]” → “mat”

 Analogy: Like filling blanks in a cloze test — model must see both sides of context.

Pre-training Task 2 – Next Sentence Prediction (NSP)

Learns relationships between two sentences.

Sentence A	Sentence B	Label
The man went to the store.	He bought milk.	IsNext 
The man went to the store.	The dog barked loudly.	NotNext 

- 50% of pairs are actual next sentences
- [CLS] vector predicts IsNext / NotNext
- Improves QA and entailment performance.

Experimental Setup

Datasets & Evaluation

Pre-training: BooksCorpus + Wikipedia

Fine-tuning:

- GLUE → general understanding
- SQuAD → question answering
- SWAG → commonsense inference

Metrics: Accuracy, F1, Exact Match

Baselines: ELMo, GPT, BiDAF+ELMo

BookCorpus	
Size	7,000 self-published books - 985 million words
Creator(s)	University of Toronto and MIT
Date of release	2015

Results and Key Findings

Performance Highlights

- BERT_BASE > GPT on every GLUE task (+4.5%)
- **BERT_LARGE:**
 - GLUE 80.5
 - MNLI 86.7
 - SQuAD 1.1 F1 93.2
 - SQuAD 2.0 F1 83.1 (+5.1 over prior)

Formula (simplified):

$$P(\text{start}_i) = \exp(S \cdot T_i) / \sum \exp(S \cdot T_k)$$



Larger models = better transfer even on small data.

Strengths

Key Advantages

- Deep bidirectional context
- Unified architecture for many NLP tasks
- Minimal fine-tuning effort
- Scales well with size

💡 Analogy: BERT = a “language brain” easily retrained for new tasks.

Limitations

Challenges and Constraints

- High cost: 4 days on 64 TPUs
- Self-attention = $O(N^2)$ complexity
- [MASK] causes mismatch between training & inference
- NSP is too simple
- Max input length = 512 tokens

➡ Later models: **RoBERTa**, **Longformer** solved these.

Impact and Future Directions

Influence and Legacy

Paradigm shift: Pre-train → Fine-tune = new NLP norm

Inspired successors:

RoBERTa (better training)

ALBERT (parameter sharing)

XLNet (permutation-based)

ELECTRA (token replacement detection)

Adoption:

- Google Search
- Chatbots
- Translation systems

Future goals: cheaper training, longer texts, multimodal learning.

Summary of Technical Innovations

Why BERT Succeeded

- Architecture: deep bidirectional Transformer encoder
- Pre-training: MLM + NSP synergy
- Fine-tuning: unified input format
- Scalability: bigger = better
- Generalization: transferable across NLP tasks

Discussion and Q&A

Possible Questions

- 💡 Why do unidirectional models fail at QA?
- 💡 Why 80/10/10 masking?
- 💡 How does bidirectionality change context?
- 💡 Can models this powerful be cheaper to train?

Takeaway:

- 💡 BERT bridges *human-like understanding* and *machine reasoning* in text.